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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 7	
TITLE : Inheritance and Interfaces	

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Programme

class

```
class Animal{
    void eat()
    {
        System.out.println("this animal eat food ");
    }
}
class Dog extends Animal
{
    void bark()
    {
        System.out.println(" dog barks");
    }
}
public class main
{
    public static void main (String [] args)
    {
        Dog d = new Dog();
        d.eat();
        d.bark();
    }
}
```

Interface

```
interface Animal
{
    void sound(); // abstract method
}
class Dog implements Animal
{
    public void sound()
    {
        System.out.println("dog barks ");
    }
}
class chat implements Animal
{
    public void sound()
    {
        System.out.println("chat crie ");
    }
}
class matou extends chat
{
    public void sound()
    {
        System.out.println("matoubou  crie ");
    }
}
public class Interface
{
    public static void main(String[] args)
    {
        Animal a = new Dog();
        Animal b = new chat();
        chat m = new matou();
        a.sound();
        b.sound();
        m.sound();
    }
}
```

Output

```
PS C:\CS-JAVA> javac main.java
PS C:\CS-JAVA> java main
this animal eat food
dog barks
PS C:\CS-JAVA> javac Interface.java
PS C:\CS-JAVA> java Interface
dog barks
chat crie
matoubou  crie
PS C:\CS-JAVA> █
```

Conclusion

In conclusion, **interfaces and simple classes with constructors** both help in implementing object-oriented programming concepts in Java. An interface defines methods that classes must implement, allowing **abstraction and polymorphism**. In the example, Dog, chat, and matou provide different implementations of sound(). A simple class with a constructor, on the other hand, is mainly used to **initialize objects when they are created**. Thus, interfaces provide a common structure for different classes, while constructors initialize the data of each object.